Aerodynamics Tutorials:

1. Consider the supersonic flow over a **5° half-angle wedge** at zero angle of attack. The freestream Mach number ahead of the wedge is 2.0, and the freestream pressure and density are **1.01×10^5 N/m^2 and 1.23 kg/m^3**, respectively. The pressure on the upper and lower surfaces of the wedge are constant with **distance s** and equal to each other, namely, **$P_u = P_l = 1.31\times10^5$ N/m^2**. The pressure exerted on the base of the wedge is equal to P_∞. The shear stress varies over both the upper and lower surfaces as $\boldsymbol{\tau_w = 431 s^{-0.2}}$. The chord length, c, of the wedge is **2 m**. Calculate the drag coefficient for the wedge.

   Refer to John D. Anderson, Fundamentals of Aerodynamics or class notes

2. Consider an infinitely thin flat plate with a 1 m chord at an angle of attack of 10° in a supersonic flow. The pressure and shear stress distribution on the upper and lower surfaces are given by $\boldsymbol{P_u = 4\times10^4(x-1)^2 + 5.4\times10^4}$, $\boldsymbol{P_l = 2\times10^4(x-1)^2 + 1.73\times10^5}$, $\boldsymbol{\tau_u = 288x^{-0.2}}$, and $\boldsymbol{\tau_l = 731x^{-0.2}}$, respectively, where x is the distance from the leading edge in meters and P and τ are in Newtons per square meter. Calculate the normal and axial forces, the lift and drag, moment about the leading edge. Also, calculate the location of the center of pressure.

   Solution:

   For a flat plate at θ = 0°

$$N' = \int_0^c (p_l - p_u)\,dx$$

$$A' = \int_0^c (\tau_l - \tau_u)\,dx$$

$$L' = N'\cos\alpha - A'\sin\alpha$$

$$D' = N'\sin\alpha + A'\cos\alpha$$

$$M'_{LE} = \int_0^c (p_u - p_l)\,x\,dx$$

$$M'_{c/4} = M'_{LE} + L'(c/4)$$

$$N' = \int_0^1 \left(-2\times10^4 (x-1)^2 + 1.19\times10^5\right)dx$$

$$N' = -2\times10^4\left(\frac{x^3}{3} - x^2 + x\right)_0^1 + \left(1.19\times10^5 x\right)_0^1 = 1.12\times10^5\ \text{N}$$

$$A' = \int_0^1 \left(731x^{-0.2} + 288x^{-0.2}\right)dx = \left(1274x^{0.8}\right)_0^1 = 1274\ \text{N}$$

$$L' = 1.12\times10^5 \times \cos10^\circ - 1274\times\sin10^\circ = 1.105\times10^5\ \text{N}$$

$$D' = 1.12\times10^5 \times \sin10^\circ + 1274\times\cos10^\circ = 2.07\times10^4\ \text{N}$$

$$M'_{LE} = \int_0^1 \left(2\times10^4 (x-1)^2 - 1.19\times10^5\right)x\,dx$$

$$M'_{LE} = 2\times10^4\left(\frac{x^4}{4} - 2\frac{x^3}{3} + \frac{x^2}{2}\right)_0^1 - \left(0.595\times10^5 x^2\right)_0^1 = -5.78\times10^4 \text{ Nm}$$

$$M'_{c/4} = -5.78\times10^4 + 1.105\times10^5(0.25) = -3.02\times10^4 \text{ Nm}$$

$$x_{cp} = \frac{-M'_{LE}}{N'} = -\frac{\left(-5.78\times10^4\right)}{1.12\times10^5} = 0.516 \text{ m}$$

3. A steady, incompressible, two-dimensional velocity is given by

$$\vec{V} = (u, v) = (0.5 + 0.8x)\hat{\imath} + (1.5 - 0.8y)\hat{\jmath}$$

Where the x- and y-coordinates are in meters and the magnitude of velocity is in m/s. A stagnation point is defined as a point in the flow field where the velocity is zero. Determine if there are any stagnation points in this flow field and, if so where?

Solution:

At the stagnation point,

$$\vec{\mathbf{V}} = \mathbf{0}$$

If $\vec{V} = 0$, then u = 0 and v = 0

$$\mathbf{u = (0.5 + 0.8x) = 0 \Rightarrow x = -0.625\ m}$$

$$\mathbf{v = (1.5 - 0.8y) = 0 \Rightarrow y = 1.875\ m}$$

4. Consider the following steady, two-dimensional velocity field:

$$\vec{V} = (u, v) = (0.66 + 2.1x)\hat{\imath} + (-2.7 - 2.1y)\hat{\jmath}$$

Calculate the location of stagnation point.

Ans: x = -0.314 and y = -1.29

5. Consider the following steady, two-dimensional velocity field:

$$\vec{V} = (u, v) = (a^2 - (b - cx)^2)\hat{\imath} + (-2cby + 2c^2xy)\hat{\jmath}$$

Calculate the location of stagnation point.

Ans: x = (b-a)/c and y = 0

6. What is the *Eulerian description* of fluid motion? How does it differ from the Lagrangian description?

In the Eulerian description of fluid motion, we are concerned with field variables, such as velocity, pressure, temperature, etc., as functions of space and time within a flow domain or control volume. In contrast to the Lagrangian

method, fluid flows into and out of the Eulerian flow domain, and we do not keep track of the motion of particular identifiable fluid particles. The Eulerian method of studying fluid motion is not as "natural" as the Lagrangian method since the fundamental conservation laws apply to moving particles, not to fields.

7. Is the Lagrangian method of fluid flow analysis more similar to study of a system or a control volume? Explain.

   The Lagrangian method is more similar to system analysis (i.e., closed system analysis). In both cases, we follow a mass of fixed identity as it moves in a flow. In a control volume analysis, on the other hand, mass moves into and out of the control volume, and we don't follow any particular chunk of fluid. Instead, we analyze whatever fluid happens to be inside the control volume at the time. In fact, the Lagrangian analysis is the same as a system analysis in the limit as the size of the system shrinks to a point.

8. A stationary probe is placed in a fluid flow and measures pressure and temperature as functions of time at one location in the flow (Fig. 01). Is this a Lagrangian or an Eulerian measurement? Explain.

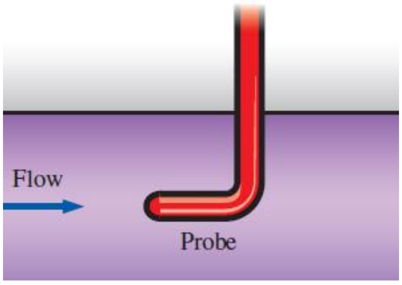


Fig.01

   Since the probe is fixed in space and the fluid flows around it, we are not following individual fluid particles as they move. Instead, we are measuring a field variable at a particular location in space. Thus, this is a Eulerian measurement. If a neutrally buoyant probe were to move with the flow, its results would be Lagrangian measurements – following fluid particles.

9. A weather balloon is launched into the atmosphere by meteorologists. When the balloon reaches an altitude where it is neutrally buoyant, it transmits information about weather conditions to monitoring stations on the ground shown in Fig.02. Is this a Lagrangian or an Eulerian measurement? Explain.

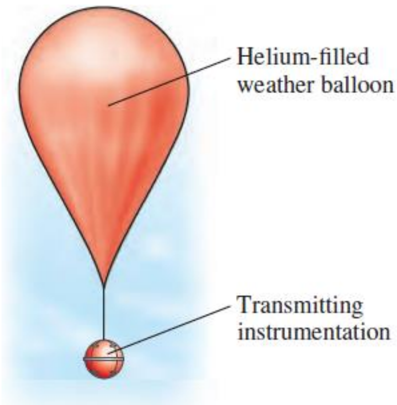


Fig. 02

Since the weather balloon moves with the air and is neutrally buoyant, we are following individual "fluid particles" as they move through the atmosphere. Thus, this is a Lagrangian measurement. Note that in this case the "fluid particle" is huge, and can follow gross features of the flow – the balloon obviously cannot follow small scale turbulent fluctuations in the atmosphere. When weather monitoring instruments are mounted on the roof of a building, the results are Eulerian measurements.

10. Pitot-static probe can often be seen protruding from the underside of an airplane (Fig. 03). As the airplane flies, the probe measures relative wind speed. Is this a Lagrangian or an Eulerian measurement? Explain.

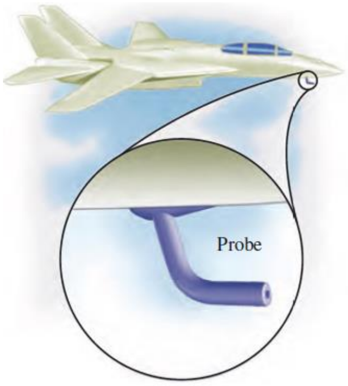


Fig. 03

Relative to the airplane, the probe is fixed and the air flows around it. We are not following individual fluid particles as they move. Instead, we are measuring a field variable at a particular location in space relative to the moving airplane. Thus, this is a Eulerian measurement. The airplane is moving, but it is not moving with the flow.

11. Find the streamline equation for the given flow field condition:

$$\vec{V} = (u, v) = x\hat{\imath} + x(x-1)(y+1)\hat{\jmath}$$

Solution:

$$\frac{dx}{u} = \frac{dy}{v} \Rightarrow \frac{dx}{x} = \frac{dy}{x(x-1)(y+1)}$$

$$\frac{x(x-1)dx}{x} = \frac{dy}{(y+1)} \Rightarrow (x-1)dx = \frac{dy}{(y+1)}$$

Integrating on both sides, we get

$$\int (x-1)dx = \int \frac{dy}{(y+1)}$$

$$\frac{x^2}{2} - x + C_1 = \ln(y+1) + C_2$$

$$\frac{x^2}{2} - x = \ln(y+1) + C_2 - C_1 = \ln(y+1) + C$$

At x = 0 and y = 0,

$$\frac{0^2}{2} - 0 = \ln(0+1) + C \Rightarrow C = 0$$

$$\frac{x^2}{2} - x = \ln(y+1) + 0$$

$$e^{\left(\frac{x^2}{2}-x\right)} = y + 1$$

The streamline equation is

$$y = e^{\left(\frac{x^2}{2}-x\right)} - 1$$

12. Find the streamline equation for the given flow field condition:

$$\vec{V} = (u, v) = (0.5 + 0.8x)\hat{\imath} + (1.5 - 0.8y)\hat{\jmath}$$

Solution:

$$\frac{dy}{dx} = \frac{1.5 - 0.8y}{0.5 + 0.8x}$$

$$\frac{dy}{1.5 - 0.8y} = \frac{dx}{0.5 + 0.8x}$$

Integrating on both sides, we get

$$\int \frac{dy}{1.5 - 0.8y} = \int \frac{dx}{0.5 + 0.8x}$$

$$-\frac{1}{0.8}\ln|1.5 - 0.8y| + C_1 = \frac{1}{0.8}\ln|0.5 + 0.8x| + C_2$$

$$-\frac{1}{0.8}\ln|1.5 - 0.8y| - \frac{1}{0.8}\ln|0.5 + 0.8x| = C_2 - C_1 = C$$

$$\ln|1.5 - 0.8y| + \ln|0.5 + 0.8x| = -0.8 * C$$

$$\ln(C) = \text{Constant} = C$$

$$\ln|1.5 - 0.8y| + \ln|0.5 + 0.8x| = -0.8 * \ln C$$

$$\ln|(1.5 - 0.8y)(0.5 + 0.8x)| = \ln C^{-0.8}$$

$$\ln|(1.5 - 0.8y)(0.5 + 0.8x)| = \ln C_0$$

$$(1.5 - 0.8y)(0.5 + 0.8x) = C_0$$

$$1.5 - 0.8\mathrm{y} = \frac{\mathrm{C_0}}{0.5 + 0.8\mathrm{x}}$$

$$\mathrm{y} = \frac{\mathrm{C_0}}{0.8 * (0.5 + 0.8\mathrm{x})} + 1.875, \textbf{Taking } (-\mathrm{C_0}) = \mathrm{C_0}$$

$$\mathrm{y} = \frac{\mathrm{C_0}}{0.8 * (0.5 + 0.8\mathrm{x})} + 1.875$$

13. The velocity field for a flow is given by

$$\vec{\mathrm{V}} = \mathrm{u}\hat{\mathrm{i}} + \mathrm{v}\hat{\mathrm{j}} + \mathrm{w}\hat{\mathrm{k}}$$

Where u = 3x, v = -2y and w = 2z. Find the streamline that will pass through the point (1, 1, 0).

Solution:

$$\frac{\mathrm{dx}}{\mathrm{u}} = \frac{\mathrm{dy}}{\mathrm{v}} = \frac{\mathrm{dz}}{\mathrm{w}}$$

$$\frac{\mathrm{dx}}{3\mathrm{x}} = \frac{\mathrm{dy}}{-2\mathrm{y}} = \frac{\mathrm{dz}}{2\mathrm{z}}$$

For the first two pairs, we have

$$\frac{\mathrm{dx}}{3\mathrm{x}} = \frac{\mathrm{dy}}{-2\mathrm{y}}$$

Integrating on both sides, we get

$$\frac{1}{3}\ln x + \ln \mathrm{C_1} = -\frac{1}{2}\ln y + \ln \mathrm{C_2}$$

$$\frac{1}{3}\ln x = -\frac{1}{2}\ln y + \ln \mathrm{C_2} - \ln \mathrm{C_1} = -\frac{1}{2}\ln y + \ln \mathrm{C}$$

$$x^{1/3}y^{-1/2} = \mathrm{C}$$

For the point given x = 1 and y = 1

$$1^{1/3}1^{1/2} = \mathrm{C} = 1$$

$$x^{1/3}y^{-1/2} = 1$$

$$\mathrm{y} = \mathrm{x}^{2/3}$$

On the other hand,

$$\frac{\mathrm{dx}}{3\mathrm{x}} = \frac{\mathrm{dz}}{2\mathrm{z}}$$

Integrating on both the sides, we get C = 0 and z = 0

The streamline is given by,

$$y = x^{2/3}; z = 0$$

14. A steady, incompressible, two-dimensional velocity field is given by,

$$\vec{V} = (u, v) = (4.35 + 0.656x)\hat{i} + (-1.22 - 0.656y)\hat{j}$$

Generate an analytical expression for the flow streamlines.

Solution:

$$\mathbf{y = \frac{C}{0.656(4.35 + 0.656x)} - 1.85796}$$

15. **Pathline:** A pathline is the actual path travelled by an individual fluid particle over some time period. It indicates the exact route along which a fluid particle travels from its starting point to its ending point. Unlike streamlines, pathlines are not instantaneous, but involve a finite time period. If a flow field is steady, streamlines, pathlines, and streaklines are identical.

16. **Streamline:** A streamline is a curve that is everywhere tangent to the instantaneous local velocity vector. It indicates the instantaneous direction of fluid motion throughout the flow field. If a flow field is steady, streamlines, pathlines, and streaklines are identical.

17. **Streakline:** A streakline is the locus of fluid particles that have passed sequentially through a prescribed point in the flow. Streaklines are very different than streamlines. Streamlines are instantaneous curves, everywhere tangent to the local velocity, while streaklines are produced over a finite time period. In an unsteady flow, streaklines distort and then retain features of that distorted shape even as the flow field changes, whereas streamlines change instantaneously with the flow field.

18. Consider the visualization of flow over a sphere in Fig. 05. Are we seeing streamlines, streaklines, or pathlines? Explain.

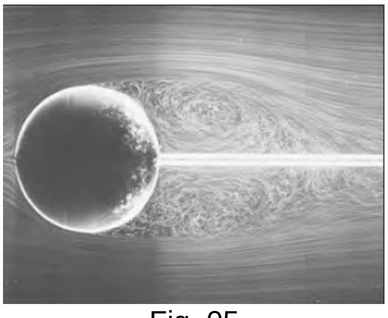


Fig. 05

Since the picture is a snapshot of smoke streaks in air, each streak shows the time history of smoke that was introduced earlier from the smoke wire. Thus these are streaklines. Since the flow appears to be unsteady, these streaklines are not the same as pathlines or streamlines. It is assumed that the smoke follows the flow of the air. If the smoke is neutrally buoyant, this is a reasonable assumption. In actuality, the smoke rises a bit since it is hot; however, the air speeds are high enough that this effect is negligible.

19. Consider the velocity field of an unsteady, two-dimensional flow given by

$$\vec{V} = (u, v) = (0.5 + 0.8x)\hat{\imath} + (1.5 + 2.5\sin(\omega t) + 0.8y)\hat{\jmath}$$

Where angular frequency ω is equal to 2π rad/s (a physical frequency of 1 Hz). Verify that this flow field can be approximated as incompressible.

Solution:

From the continuity equation,

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = -\frac{\partial \rho}{\partial t}$$

For an incompressible flow,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

For 2D flow,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(0.5 + 0.8x) = 0.8$$

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y}(1.5 - 0.8y) = -0.8$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.8 - 0.8 = 0$$

20. Consider a steady velocity field given by

$$\vec{V} = (u, v, w) = a(x^2y + y^2)\hat{\imath} + bxy^2\hat{\jmath} + cx\hat{k}$$

Where a, b, and c are constants. Under what condition is this flow field incompressible?

Solution:

From the continuity equation,

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = -\frac{\partial \rho}{\partial t}$$

For an incompressible flow,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

For 2D flow,

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0=2axy+2bxy$$

Thus, to guarantee incompressibility, constants *a* and *b* must be equal in magnitude but opposite in sign.

***Condition for incompressibility*:**

a = -b

21. A steady, two-dimensional, incompressible flow field in the xy-plane has a stream function given by $\mathbf{\Psi=ax^3+by+cx}$, where a, b, and c are constants: a = 0.5 m/s, b = -2.0 m/s, and c = -1.5 m/s. Obtain expressions for velocity components u and v and derive the streamline equation from the given conditions.

To find the streamline equation, solve the given equation for y as a function of x and Ψ.

Solution:

To obtain expressions for u and v, differentiate the given stream function

$$u=\frac{\partial\Psi}{\partial y}=b;v=-\frac{\partial\Psi}{\partial x}=-3ax^2-c$$

Since u is not a function of x and v is not a function of y, continuity equation is satisfied.

Streamline equation:

$$y=\frac{\Psi-ax^3-cx}{b}$$

A steady, two-dimensional, incompressible flow field in the *xy*-plane has the following stream function: $\Psi=ax^2+bxy+cy^2$, where *a*, *b*, and *c* are constants. (*a*) Obtain expressions for velocity components *u* and υ. (*b*) Verify that the flow field satisfies the incompressible continuity equation.

Solution:

Assumptions: 1. The flow is steady. **2.** The flow is incompressible (this assumption is to be verified). **3.** The flow is two-dimensional in the *x-y* plane, implying that *w* = 0 and neither *u* nor *v* depend on *z*.

$$u=\frac{\partial\Psi}{\partial y}=bx+2cy;v=-\frac{\partial\Psi}{\partial x}=-2ax-by$$

Continuity equation for incompressible flow,

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=b-b=0$$

22. A steady, two-dimensional, incompressible flow field in the *xy*-plane has the following stream function: $\Psi = ax^2 - by^2 + cx + dxy$, where *a*, *b*, c, and d are constants. (*a*) Obtain expressions for velocity components *u* and v. (*b*) Verify that the flow field satisfies the incompressible continuity equation.

Solution:

Assumptions: 1. The flow is steady. **2.** The flow is incompressible (this assumption is to be verified). **3.** The flow is two-dimensional in the *x*-*y* plane, implying that *w* = 0 and neither *u* nor *v* depend on *z*.

$$u = \frac{\partial \Psi}{\partial y} = -2by + dx; v = -\frac{\partial \Psi}{\partial x} = -2ax - c - dy$$

Continuity equation for incompressible flow,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = d - d = 0$$

23. Consider a steady, two-dimensional, incompressible flow field with $\mathbf{u} = \mathbf{ax} + \mathbf{b}$ and $\mathbf{v} = -\mathbf{ay} + \mathbf{cx}$ where a, b, and c are constants: a = 0.5 m/s, b = 1.5 m/s, and c = 0.35 m/s. Generate an expression for the stream function and streamline.

Solution:

To obtain expressions for u and v, differentiate the given stream function

$$u = \frac{\partial \Psi}{\partial y} = ax + b$$

Integrate with respect to y, we get

$$\Psi = axy + by + g(x)$$

Noting that this is a *partial* integration, so we add an arbitrary function g(x) of the other variable, *x*, rather than a constant of integration.

We know that

$$v = -\frac{\partial \Psi}{\partial x} = \frac{\partial}{\partial x}(axy + by + g(x)) = -ay - g'(x)$$

where $g'(x)$ denotes dg/dx since g is a function of only one variable x.

$$v = -ay - g'(x) \text{ and } \mathbf{v} = -\mathbf{ay} + \mathbf{cx}$$

Comparing the two expressions, we get

$$g'(x) = -\mathbf{cx}$$

Integrating with respect to x, we get

$$g(x) = -c\frac{x^2}{2} + \text{Constant}$$

The final expression for Ψ (**stream function**)

$$\Psi = axy + by - c\frac{x^2}{2} + \text{Constant}$$

Equation for streamline,

$$y = \frac{\Psi + cx^2/2}{ax + b}$$

24. Consider the steady, two-dimensional, incompressible velocity field given as,

$$\vec{V} = (u, v) = (ax + b)\hat{\imath} + (-ay + cx)\hat{\jmath}$$

Calculate the pressure as function of x and y.

Assumptions **1.** The flow is steady and incompressible. **2.** The fluid has constant properties. **3.** The flow is two-dimensional in the *xy*-plane. **4.** Gravity does not act in either the *x*- or *y*-direction.

Solution:

First, we check whether the given velocity field satisfies the two-dimensional, incompressible continuity equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = a - a = 0$$

If continuity were *not* satisfied, we would stop our analysis, the given velocity field would not be physically possible, and we could not calculate a pressure field.

Consider the *y*-component of the Navier–Stokes equation:

$$\boldsymbol{\rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}\right) = -\frac{\partial P}{\partial y} + \rho a_x + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right)}$$

$$\frac{\partial v}{\partial t} = 0, \textbf{Flow is steady}$$

$$u\frac{\partial v}{\partial x} = (ax + b)c;\ v\frac{\partial v}{\partial x} = (-ay + cx)(-a)$$

$$w\frac{\partial v}{\partial z} = \frac{\partial^2 v}{\partial z^2} = 0$$

$$\rho a_x = 0, \textbf{No body force}$$

$$\frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 v}{\partial y^2} = 0$$

The y-momentum equation reduces to,

$$\frac{\partial P}{\partial y} = \rho(-acx - bc - a^2y + acx) = \rho(-bc - a^2y)$$

Similarly,

$$\frac{\partial P}{\partial x} = \boldsymbol{\rho(-a^2x - ab)}$$

In order for a steady flow solution to exist, *P* cannot be a function of time. Furthermore, a physically realistic steady, incompressible flow field requires a pressure field *P* (*x*, *y*) that is a smooth function of *x* and *y* (there can be no sudden discontinuities in either *P* or a derivative of *P*).

P(x, y) is a smooth function of x and y only if the order of differentiation does not matter (**cross-Differentiation**).

$$\boldsymbol{\frac{\partial^2 P}{\partial x \partial y} = \frac{\partial^2 P}{\partial y \partial x} (\mathbf{cross - Differentiation, xy - plane})}$$

For a two-dimensional flow field in the *xy*-plane, **cross-differentiation** reveals whether pressure *P* is a smooth function.

$$\frac{\partial^2 P}{\partial x\, \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial P}{\partial y}\right) = 0;\ \frac{\partial^2 P}{\partial y\, \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial P}{\partial x}\right) = 0$$

Pressure field from y-momentum,

$$P(x, y) = \rho\left(-bcy - \frac{a^2y^2}{2}\right) + g(x)$$

$g(x)$ is an arbitrary function of the other variable x rather than a constant of integration since this is a partial integration.

Partial derivative with respect to x to obtain

$$\frac{\partial P}{\partial x} = g'(x) = \boldsymbol{\rho(-a^2x - ab)}$$

We now integrate the above equation to obtain the function g(x):

$$g(x) = \rho\left(-\frac{a^2x^2}{2} - abx\right) + \text{Constant}$$

$$P(x, y) = \rho\left(-\frac{a^2x^2}{2} - \frac{a^2y^2}{2} - abx - bcy\right) + \text{Constant}$$

25. Consider the steady, two-dimensional, incompressible velocity field given as,

$$\vec{V} = (u, v) = (-ax^2)\hat{\imath} + (2axy)\hat{\jmath}$$

Calculate the pressure as function of x and y.

Assumptions 1. The flow is steady and incompressible. **2.** The fluid has constant properties. **3.** The flow is two-dimensional in the *xy*-plane. **4.** Gravity does not act in either the *x*- or *y*-direction.

Solution:

$$\frac{\partial P}{\partial x} = - - 2\rho a^2 x^3 - 2\mu a; \ \frac{\partial P}{\partial y} = -2\rho a^2 x^2 y$$

Since the cross-derivative terms do not match, *P* is *not* a smooth function of *x* and *y*. Thus, **we are unable to calculate a steady, incompressible, two-dimensional pressure field with the given velocity field**. We cannot proceed any further.

26. What are the three major assumptions used in the derivation of the Bernoulli equation?

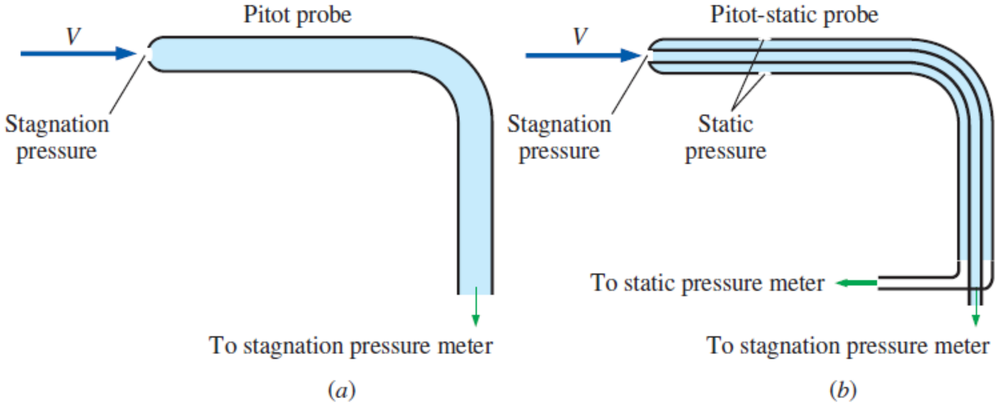
The three major assumptions used in the derivation of the Bernoulli equation are that the flow is steady, there is negligible frictional effects, and the flow is incompressible. If any one of these assumptions is not valid, the Bernoulli equation should not be used. Unfortunately, many people use it anyway, leading to errors.

27. What is stagnation pressure? Explain how it can be measured.

The sum of the static and dynamic pressures is called the stagnation pressure, and it is expressed as

$$\mathbf{P_{stag} = P_{static} + \frac{1}{2}\rho V^2}$$

The stagnation pressure can be measured by a Pitot tube whose inlet is normal to the flow. Stagnation pressure, as its name implies, is the pressure obtained when a flowing fluid is brought to rest Isentropically, at a so-called stagnation point.



(*a*) A **Pitot probe** measures stagnation pressure at the nose of the probe, while (*b*) a **Pitot-static probe** measures both stagnation pressure and static pressure, from which the flow speed is calculated.

28. What is significant about curves of constant stream function? Explain why the stream function is useful in fluid mechanics.

The difference in the value of ψ from one streamline to another is equal to the volume flow rate per unit width between the two streamlines. This fact about the stream function can be used to calculate the volume flow rate in certain applications.

29. A **boundary layer** is a thin region near a wall in which viscous (frictional) forces are very important due to the no-slip boundary condition. The steady, incompressible, two-dimensional, boundary layer developing along a flat plate aligned with the free-stream flow is sketched in Fig. 06. The flow upstream of the plate is uniform, but boundary layer thickness δ grows with x along the plate due to viscous effects. Sketch some streamlines, both within the boundary layer and above the boundary layer. Is δ(x) a streamline? (Hint: Pay particular attention to the fact that for steady, incompressible, two-dimensional flow the volume flow rate per unit width between any two streamlines is constant.)

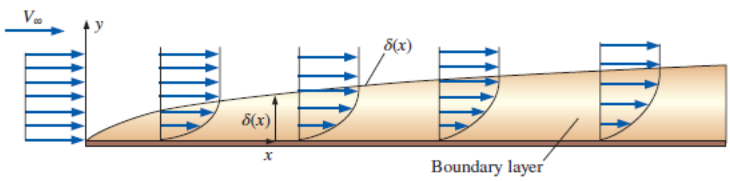


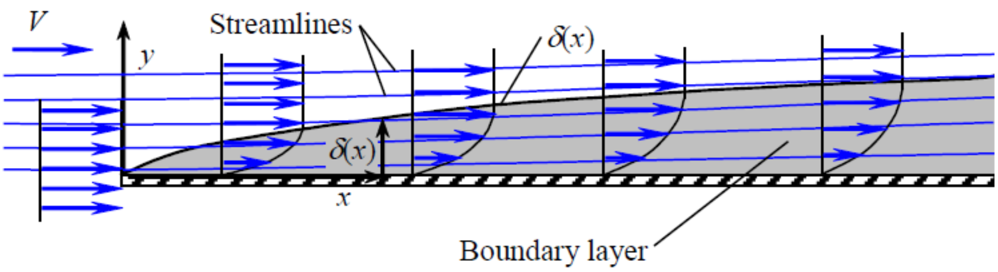
Fig. 06

Solution:

- **Sketch some streamlines for boundary layer flow.**

Assumptions: 1. The flow is steady. **2.** The flow is incompressible. **3.** The flow is two-dimensional in the x-y plane.

Since there is no flow reversal, we can be sure that $\delta(x)$ is not a streamline. In fact, streamlines must *cross* $\delta(x)$. Furthermore, at any given *y* location above the plate the fluid speed decreases as the boundary layer grows downstream. Hence, *the streamlines must diverge*. The bottom line is that the streamlines veer slightly upward away from the wall to compensate for the loss of speed in the boundary layer.



Streamlines above and within a flat plate boundary layer; since streamlines cross the curve $\delta(x)$, $\delta(x)$ cannot itself be a streamline of the flow. Furthermore, streamlines within the boundary layer change direction suddenly up because of decreasing speeds within the boundary layer. As the boundary layer grows in thickness, more and more streamlines end up inside the boundary layer.